

ABU DHABI BATEEN, United Arab Emirates

WMO: 412160

Lat: 24.428N

Lon: 54.458E

Elev: 5

StdP: 101.27

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.0	15.0	1.0	4.1	31.2	3.5	4.9	30.4	9.4	20.7	8.5	21.1	1.4	150	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.6	44.7	23.4	43.1	23.7	41.9	23.9	31.0	35.0	30.5	34.7	30.0	34.5	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.1	27.4	33.8	29.2	26.0	33.6	29.0	25.6	33.6	105.3	35.0	102.9	34.4	100.2	34.4	32.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	7.9	7.0	12.0	46.9	1.0	0.9	11.3	47.5	10.8	48.1	10.2	48.6	9.5	49.3		
			8.6	32.0	1.5	0.4	7.5	32.3	6.6	32.6	5.8	32.8	4.7	33.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.1	20.1	21.2	24.0	28.4	32.9	34.7	36.8	37.1	34.5	30.7	26.0	21.8
	DBStd	6.40	1.72	2.73	3.06	3.18	2.21	1.85	2.18	1.64	1.69	1.87	2.03	1.87
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	6	3	2	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6961	314	315	433	553	709	742	829	841	734	643	481	366
	CDD18.3	3925	59	84	174	303	451	492	571	582	484	385	231	108
	CDH23.3	53644	202	459	1239	3451	6596	7697	9310	9398	7439	5239	2153	462
	CDH26.7	34400	31	152	474	1741	4207	5302	6831	6918	5045	2914	729	56
Wind	WSAvg	3.3	3.0	3.6	3.8	3.4	3.4	3.5	3.4	3.4	3.2	2.9	2.9	2.8
Precipitation	PrecAvg	60	11	11	14	6	2	1	2	2	0	0	2	9
	PrecMax	159	52	92	58	53	31	20	19	27	2	7	25	63
	PrecMin	3	0	1	0	0	0	0	0	0	0	0	0	0
	PrecStd	46	17	18	18	13	6	4	5	5	1	1	6	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	33.9	37.0	41.0	44.1	44.8	46.2	45.9	43.5	39.0	35.0	30.0
		MCWB	17.2	17.6	17.9	19.9	21.4	22.1	23.6	23.4	22.6	22.0	19.3	19.1
	2%	DB	27.2	30.8	33.9	38.8	42.0	42.8	44.9	44.2	41.4	37.2	33.0	28.1
		MCWB	17.0	17.3	18.3	20.0	21.2	22.9	23.9	23.8	23.3	22.9	20.9	19.7
	5%	DB	25.6	28.2	31.2	36.6	40.1	41.1	43.1	43.0	40.0	36.1	31.5	27.0
		MCWB	17.1	17.1	18.3	20.1	21.7	23.6	24.3	24.3	23.9	23.2	21.3	19.8
	10%	DB	24.0	26.0	29.1	34.1	38.2	39.8	41.2	41.2	38.4	35.0	30.2	25.9
		MCWB	17.4	17.4	18.4	20.0	22.1	24.0	25.3	25.0	25.0	23.7	21.4	19.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.5	21.2	22.7	25.5	28.2	30.6	31.7	31.7	30.8	29.1	25.2	22.7
		MCDB	23.7	25.2	27.3	29.9	33.1	34.7	35.6	35.2	33.9	32.5	29.4	25.6
	2%	WB	19.6	20.4	21.9	24.5	27.2	29.9	30.9	31.1	30.2	28.2	24.2	22.1
		MCDB	23.0	24.1	26.3	29.6	32.0	34.2	35.3	35.0	33.8	31.9	28.8	25.6
	5%	WB	19.0	19.6	21.2	23.5	26.4	29.2	30.3	30.6	29.6	27.5	23.6	21.2
		MCDB	22.7	23.6	25.4	29.5	31.7	33.9	34.9	34.8	33.6	31.7	28.3	24.7
	10%	WB	18.4	18.9	20.6	22.6	25.6	28.5	29.8	30.1	29.1	26.8	22.9	20.6
		MCDB	22.3	23.4	25.0	29.1	31.6	33.5	34.8	34.7	33.4	31.4	27.9	24.2
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.6	9.1	10.8	11.7	11.1	10.3	10.6	9.9	9.5	8.5	8.0
		MCDBR	11.0	12.2	12.5	13.9	14.3	13.0	12.4	12.2	11.5	10.5	9.8	9.1
		MCWBR	5.0	5.4	5.4	6.2	6.4	6.8	7.4	7.7	7.1	5.8	4.8	4.4
	5% WB	MCDBR	9.1	9.4	9.2	12.3	12.1	11.1	10.2	10.6	10.0	9.5	9.1	8.4
		MCWBR	4.8	4.9	4.8	5.8	5.9	5.9	6.3	6.5	5.9	5.1	4.5	4.4
Clear-Sky Solar Irradiance	taub	0.454	0.506	0.572	0.611	0.596	0.634	0.727	0.700	0.598	0.508	0.463	0.444	
	taud	2.035	1.891	1.740	1.689	1.706	1.669	1.555	1.606	1.755	1.930	2.050	2.068	
	Ebn at Noon	787	773	744	726	734	702	639	655	713	760	772	778	
	Edh at Noon	153	189	230	247	242	249	279	265	223	179	149	143	
All-Sky Solar Radiation	RadAvg	4.20	4.95	5.82	6.64	7.36	7.49	7.03	6.79	6.33	5.53	4.50	3.94	
	RadStd	0.23	0.39	0.39	0.40	0.20	0.15	0.23	0.14	0.13	0.09	0.11	0.24	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.50	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	+165	+160

Nomenclature: See separate page